Programs for a Frozen Machine: The Description Length of Language and Task in Optimized Prefixes

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Abstract

A frozen language model is a fixed machine; a soft prefix that makes it perform a task is a program for that machine, and the shortest such program is a model-relative description length of the task. We measure that length for a frozen Qwen2.5-1.5B base model over a grid of three tasks (translation from English, factual recall, sentence continuation) and three output languages (Russian, German, English), in two units: prefix slots and variational bits under a bits-back code. Four findings. (i) Slots are not a description length: translation saturates at one slot, and one trained slot beats the natural-language prompt everywhere. (ii) In bits, we find translation programs of about 500 bits at 10% excess loss, Russian recall programs five times that, and English recall the most expensive of all. (iii) Sharing provides no consistent compression advantage under the tested configurations, and neither nesting is consistently cheaper: a hierarchy with one centre per task and a deviation per language costs 1.9 times the independent programs on the measurable cells, and one centre per language with task deviations only breaks even on Russian and German. The asymmetry is in the mechanism: patching the first block alone recovers 70–87% of a translation program's effect but only 36–58% of a recall program's, with further recovery at deeper blocks. (iv) The variational objective selects compact programs but the standard optimiser does not find them: a conservative step size and checkpoint selection on the full objective lower the measured frontier by 20–71%.

1 Introduction

Prompt tuning [7] and prefix tuning [9] adapt a frozen language model by optimising a few continuous vectors prepended to the input. Read as engineering, these vectors are parameters. Read through the lens of algorithmic information theory they are programs: the frozen model is the interpreter, the prefix is the program, the input is the data and the target is the result. Shaw et al. [11] make this reading precise, showing that prompt tokens form a program tape for a Transformer interpreter and that the variational code length of a prompt is an asymptotically optimal description-length objective. Cosma [2] define prompting complexity as a model-relative, resource-bounded Kolmogorov complexity.

This project measures the description length of the programs for multilingual tasks. For a frozen model M, task ρ and output language l, let $K_M(\rho, l)$ be the length of the shortest prefix that performs ρ in l at a chosen distortion. We ask:

- How is $K_M(\rho, l)$ split between telling the model which task to perform and which language to produce?
- Are the two parts separable, so that a task part and a language part can be assembled?

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- How far apart are the programs for different language–task pairs, and do they cluster by task or by language?
- Where in depth does the model read each part?
- Is the measurement itself trustworthy: does gradient descent find the short programs that the objective prefers?

Our contributions are a measurement design and five results. The design (Section 3) fixes two units, slots and bits, one normalised distortion, two distortion thresholds chosen in advance from a development pilot, and a hierarchical variational prefix whose centre and deviations put task bits and language bits in the same currency. The results (Sections 5–6) are the four findings of the abstract plus a negative one: the information-distance matrix over language–task pairs has no cluster structure at one seed.

2 Related work

Description length of models and probes Hinton and van Camp [4] introduced the variational (bits-back) code for network weights; Blier and Ollivier [1] compared variational and prequential codes for deep networks and found the prequential one far shorter; Voita and Titov [13] used prequential coding to measure what a probe learns. We use both codes: variational for the rate of a prefix and prequential for the distance between prefixes.

Soft prompts as programs Shaw et al. [11] give the universality construction and the adaptive prior we build on, and report that the objective selects compressible solutions that random-initialisation training fails to find. Cosma [2] measure shortest prompts, discrete and soft, as a complexity. Vu et al. [14] treat learned prompts as task embeddings that predict transfer; Vykopal et al. [15] combine soft language prompts with task adapters. Khashabi et al. [6] show that a working soft prompt need not read as words, which is why every claim here is functional: transfer, composition, conditioning or intervention, never a nearest-token reading.

Task and language inside the model Function vectors [12] show that a task can be carried by one hidden vector patched into the residual stream. Wendler et al. [16] and Dumas et al. [3] locate a language-agnostic concept representation in the middle blocks with language resolved later; our patching probes test where a learned task part and a learned language part are read.

3 Measurement framework

Interpreter, program, distortion The prefix $P \in \mathbb{R}^{k \times d}$ occupies the first k positions of the input sequence, where $d = 1536$ is the model's embedding dimension. The model's weights, tokenizer and format are fixed. Distortion is measured by the teacher-forced code length L of the target in bits per target token. Let m be the number of examples in the dataset being evaluated, (x_n, y_n) the input and target of example n, and N the total number of target tokens. For a prefix P, define the average negative log-likelihood as

$$L(P) = -\frac{1}{N} \sum_{n=1}^{m} \log_2 p_M(y_n \mid x_n, P),$$

where the probability of a target is the product over its tokens, each conditioned on the ground-truth preceding target tokens. In the training objective below, L and N refer to the training set; reported distortion uses held-out losses.

To compare across task–language pairs, we normalise the loss per pair:

$$\hat{D} = \frac{L - L_{\text{floor}}}{L_{\text{none}} - L_{\text{floor}}}, \tag{1}$$

where L_{none} is the no-prefix loss and $L_{\text{floor}} = \min\{L_{\text{LoRA}}, L_{\text{best prefix}}\}$ is the lower of two empirical reference losses: a per-pair rank-16 LoRA fine-tune of the attention projection matrices, with the pretrained weights frozen, and the best unpenalised prefix over the k sweep. All three losses are evaluated on the same dataset. The distortion thresholds $\hat{D} \leq 0.25$ and $\hat{D} \leq 0.10$ were chosen in advance from a development pilot and $\hat{D} \leq 0.05$ is reported as exploratory.

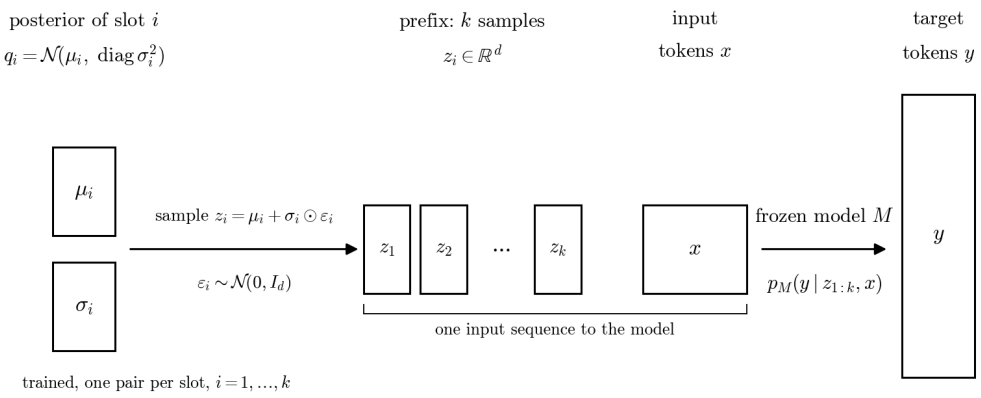


Figure 1: The sampling step of a variational prefix. Each slot i has a diagonal Gaussian posterior $q_i = \mathcal{N}(\mu_i, \mathrm{diag}\,\sigma_i^2)$ with trained μ_i and σ_i, and d is the model's embedding dimension (1536). At every forward call one sample $z_i = \mu_i + \sigma_i \odot \varepsilon_i$, with $\varepsilon_i \sim \mathcal{N}(0, I)$, is drawn per slot. The k samples form the prefix, which is prepended to the input tokens x and passed through the frozen model M.

Two units and rate–distortion objective *Slots* are the number k of prefix positions and the slot complexity k^* is the smallest k that reaches a distortion threshold, found by sweeping k with free, unpenalised vectors. *Bits* are the description length of the prefix under a prior: each slot i has a diagonal Gaussian posterior $q_i = \mathcal{N}(\mu_i, \mathrm{diag}\,\sigma_i^2)$, and the full prefix distribution is $q = \bigotimes_{i=1}^{k} q_i$, trained using the reparameterisation trick (as in VAEs; see Figure 1). We measure the expected description length of the sampled prefix, in bits, as $R(q) = D_{\mathrm{KL}}(q \,\|\, \pi_0)/\ln 2$, where π_0 is the product of independent per-slot diagonal Gaussian priors fitted to the model's token-embedding matrix (under an idealised bits-back coding scheme). The rate–distortion frontier of a task–language pair is the smallest description length among runs that reach a distortion threshold, over k and the KL weight β. Distortion of a variational prefix is scored on posterior samples, because the sample is the coded object (Section 6 shows what goes wrong when the posterior mean is scored instead).

The variational objective is

$$\mathcal{J}_\beta(q) = \mathbb{E}_{P\sim q}[L(P)] + \frac{\beta}{N}\frac{D_{\mathrm{KL}}(q\|\pi_0)}{\ln 2}.$$

Both terms are measured in bits per training target token. The first represents the expected data code length per token, while the second represents the prefix description length averaged over the training tokens. Multiplying the objective by N yields

$$N\mathcal{J}_\beta(q) = N\mathbb{E}_{P\sim q}[L(P)] + \beta R(q),$$

where $R(q) = D_{\mathrm{KL}}(q\|\pi_0)/\ln 2$ is the prefix description length. For $\beta = 1$, this corresponds to the idealised total expected description length under the assumed bits-back coding scheme, ignoring coding overheads. The expectation over $P \sim q$ is estimated with one sample per training step; reported distortion averages eight passes over the test set with a fresh sample per batch.

Hierarchy We test whether prefixes for different task–language pairs can be compressed by sharing a common component. For each group (either the languages of one task or the tasks of one language), we learn a shared centre c and a separate deviation δ_g for each member g, giving $P_g = c + \delta_g$. Both have the dimensions of an ordinary prefix, but may have different description lengths. Grouping by task shares a centre across languages and grouping by language shares one across tasks.

We learn Gaussian distributions q_c and q_g over the centre and deviations. The centre uses the original prior π_0, while each deviation uses a zero-centred Gaussian prior π_δ with the same covariance as π_0 ($\tau = 1$), so that a displacement costs the same bits in a deviation as in a whole prefix. The group's total description length is

$$R_{\mathrm{group}} = \frac{1}{\ln 2}\left(D_{\mathrm{KL}}(q_c\|\pi_0) + \sum_g D_{\mathrm{KL}}(q_g\|\pi_\delta)\right).$$

The centre is encoded only once, so sharing saves bits when this cost is smaller than that of independently encoding every member. A group reaches a distortion threshold only when every member does in the same run.

Training alternates between pairs within each group. For a pair g, we construct its prefix $P_g = c + \delta_g$ and minimise

$$\mathcal{J}_g = \mathbb{E}[L_g(c + \delta_g)] + \beta \left(\frac{R_c}{N_{\text{group}}} + \frac{R_g}{N_g} \right),$$

where R_c and R_g are the description lengths of the shared centre and the pair-specific deviation, N_g is the number of training target tokens for pair g, and $N_{\text{group}} = \sum_g N_g$.

Derived measurements We use four additional experiments to study how the learned prefixes represent tasks and languages.

Composition. We learn one prefix component per task and one per language, shared across task–language pairs. The components are concatenated to form a complete prefix. We first train the components jointly on all pairs. We then test whether they generalise to unseen combinations by excluding selected pairs from training and assembling their prefixes from components learned on other pairs: for example, exclude Russian translation from training, learn the translation component from German translation and the Russian component from Russian recall, concatenate them, and test whether the model translates into Russian without any additional training.

Distance. We measure the prequential cost $K(a \mid b)$ of training for pair a, initialised from pair b's prefix, and compare it with the from-scratch cost $K(a)$. The prequential cost is the predictive loss accumulated while learning, over a fixed budget of 512 examples in doubling blocks. The symmetric, normalised distance is

$$d(a, b) = \frac{K(a \mid b) + K(b \mid a)}{K(a) + K(b)},$$

the sum-form analogue of the normalised information distance of Li et al. [8], with prequential learning cost in place of Kolmogorov complexity. We also apply each trained prefix unchanged to other pairs to measure zero-shot transfer.

Mechanism. We investigate which transformer blocks use information from the learned prefix. We run the model with a trained prefix and record its activations, then run it with a random prefix of the same k. At one selected block, we replace only the activations at the prefix positions, at the input to self-attention, with those recorded from the trained-prefix run; all other activations remain unchanged. We repeat this for each block separately and measure how much of the trained prefix's improvement in loss each replacement recovers. Replacing at all blocks at once recovers it exactly.

Input compression (exploratory, Appendix C). We replace each input with a small number of trainable vectors specific to that example, and compare the resulting loss with and without a learned task prefix.

4 Experimental setup

Model, data, grid Qwen2.5-1.5B base [10], frozen, bfloat16 on an A100. Translation and continuation use FLORES+ (dev split for training and selection, devtest for reporting); recall uses mLAMA [5] with an English cloze prefix and the object rendered in the target language, split by item. Translation into English is excluded, so the grid has eight cells: `translation-{ru,de}`, `recall-{ru,de,en}`, `continuation-{ru,de,en}`. Continuation is unmeasurable: its floor lies within 0.10–0.16 bits per token of the no-prefix loss, below the seed variation, so it is kept in the tables and excluded from every claim.

Runs Design A (free, unpenalised prefixes): $k \in \{1, 2, 4, 8, 16, 32\}$, two seeds, 1500 steps, batch 16, AdamW with cosine decay; repeated with the target language given, the task given, and the natural-language prompt with no prefix; per-cell LoRA floors. Flat variational: $k \in \{2, 4, 8\}$, $\beta \in \{0.3, 1, 3\}$. Hierarchical: both nestings, $k \in \{2, 4, 8\}$, $\beta \in \{0.3, 1, 3\}$. Composition: 1+1, 2+2, 4+4 slots with five hold-out sets. Distance: every cell's $k = 4$ prefix toward every cell. Mechanism: eight cells, five probes, 28 blocks. Language groups retrained on their two measurable members (translation, recall) on the same grid. Zero-rate baselines: eight draws from π_0 and eight random token rows per cell at $k \in \{2, 4\}$, scored without training. In total about 37 A100 hours (Appendix E).

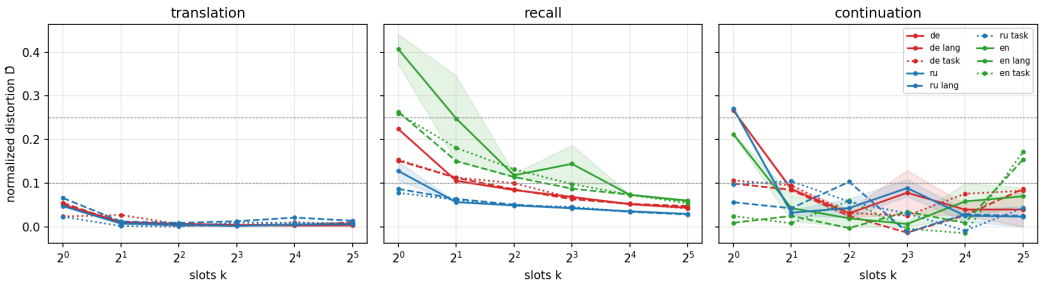


Figure 2: Normalised distortion against the number of slots, Design A, seed means with the seed range shaded. Dashed lines mark the thresholds; line style is the conditioning format. Continuation (right) is noise.

cell	$k^*(0.25)$	$k^*(0.10)$	$k^*(0.05)$, expl.
translation-ru	1	1	1
translation-de	1	1	2
recall-ru	1	2	4
recall-de	1	4	32
recall-en	2	16	not reached

Table 1: Slot complexity: the smallest k whose seed-mean $\hat{D}$ reaches the level.

Two variational protocols The preregistered protocol used learning rate 0.05, the checkpoint with the lowest development distortion at the posterior mean, and σ initialised at five times the prior. Its training histories (Section 6) showed that this overshoots the rate and selects on the wrong statistic. The corrected protocol uses learning rate 0.01, scores development distortion on samples, selects the checkpoint with the lowest objective $\mathcal{J}_\beta$, and initialises σ at the prior. It was run on the five measurable cells at $k \in \{2, 4\}$, $\beta \in \{1, 3\}$ and is used for every comparison below. Both hierarchical nestings were rerun under it, and the corrected runs are used for every hierarchical comparison.

Preregistration The hypotheses of Appendix A, the distortion thresholds, the run grid and the checkpoint rule were fixed in a written plan before the main runs; the deviations from that plan are reported in Section 7.

5 Results

5.1 Slots are not a description length

Figure 2 shows $\hat{D}$ against k. Translation reaches $\hat{D} \leq 0.10$ with one slot in every language and conditioning and is flat from there. Recall keeps improving to $k = 32$ on Russian and German with the budget binding: a 3000-step run on `recall-ru` at $k = 8$ moves $\hat{D}$ from 0.043 to 0.032 and is still falling. Seed gaps are below 0.011 on translation, 0.042 on `recall-ru` and `recall-de`, and 0.20 on `recall-en`. Table 1 gives k^*.

Two observations make the slot count unusable as a description length. First, the natural-language prompt, scored as a context with no prefix, is far worse than one trained slot: on recall it leaves $\hat{D}$ at 0.91–0.96 where one slot leaves 0.13–0.41; on translation 0.12–0.20 against 0.05. A single trained vector does more than an instruction of many tokens, so a slot count is not commensurable with a token count. Second, conditioning saves little on the slot axis: giving the language or the task changes $k^*(0.10)$ in none of the translation cells and in two recall cells (`recall-ru` 2 to 1, `recall-en` 16 to 8, for either conditioning). The savings Δk_{lang} and Δk_{task} average 1.8 slots each over the measurable cells, with 8 of those 9 slots from `recall-en`. The slot axis cannot distinguish telling the model the language from telling it the task; the bit axis can.

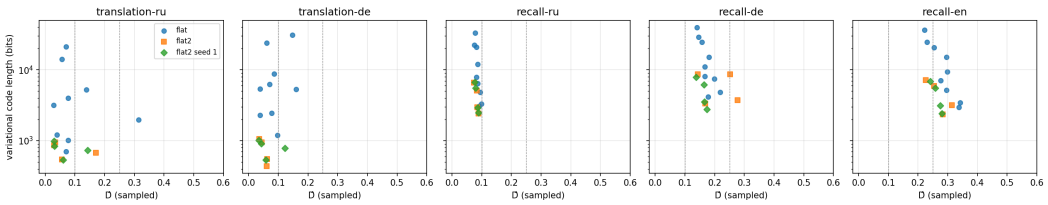


Figure 3: Rate against distortion for every variational run, per cell. Circles: preregistered protocol; squares: corrected protocol. Distortion is scored on posterior samples.

	corrected		best	prereg.
cell	$\hat{D} \leq 0.10$	$\hat{D} \leq 0.25$	$\hat{D}$	$\hat{D} \leq 0.25$
translation-de	436	436	0.036	1,194
translation-ru	542	542	0.030	699
recall-ru	2,419	2,419	0.075	3,101
recall-de	—	3,343	0.143	4,168
recall-en	—	7,155	0.226	24,686

Table 2: Rate–distortion frontier: the smallest code length in bits among runs reaching each threshold, under the corrected protocol, with the best distortion any corrected run reaches and the preregistered protocol's value at 0.25. Runs qualify by their test distortion (Section 7).

5.2 Description length in bits

Table 2 gives the frontier. The shortest translation program found costs about 500 bits, and the shortest Russian recall program five times that; German and English recall do not reach 10% excess loss within the budget, and the shortest programs found at 25% cost 3,300 and 7,200 bits. All of these are upper bounds on the shortest achievable codes (Section 7). A second seed reproduces the corrected column within 4% on three cells and within 22% on the other two, with the same ordering (Table 8). Selecting the qualifying runs on development rather than test distortion reproduces every value in the table except `recall-en` at 0.25, where the cheapest development-qualifying run (5,933 bits) has test distortion 0.253 and misses the test threshold; the 7,155-bit program is the one that qualifies on both sets. Distortion is normalised relative to the no-prefix loss, and the zero-rate program $q = \pi_0$ performs only marginally better than that baseline: k vectors drawn from π_0 leave $\hat{D}$ at 0.91–0.96 on translation and 0.93–0.98 on Russian and German recall, and make English recall worse than no prefix (1.07–1.12), averaged over eight draws (Appendix D).

The ordering across tasks within a language follows the learning gap $L_{\text{none}} - L_{\text{floor}}$ (recall 3.5–6.2 bits per token, translation 0.6–0.9), as hypothesised. The ordering across languages does not follow exposure: English recall is the dearest program, not the cheapest, in slots and in bits, and the result survives restricting the three recall test sets to the 4,212 items they share ($\hat{D}$ at $k = 4$: ru 0.049, de 0.079, en 0.122). The English LoRA floor is also the worst of the three (2.58 bits per token against 1.10 for Russian). This is consistent with the difficulty lying in the English rendering of the mLAMA items rather than in English as an output language, but tokenisation and the output distribution could contribute, and the experiments do not separate them.

5.3 Neither nesting is cheaper, and sharing gives no consistent saving

Table 3 compares each group's R_{group} with the sum of its members' independent frontiers at $\hat{D} \leq 0.25$, the only level every measurable cell reaches; continuation is excluded on both sides. Sharing provides no consistent compression advantage under the tested configurations: the one measured saving, the German language group at 0.97, is too small to be reliable with one seed. Over the five measurable cells the task-centred hierarchy costs 1.87 times the independent programs: parity on translation and 1.9 times on recall. The language-centred hierarchies, retrained on their measurable members, are at parity (1.14 and 0.97) and do not bring English recall under 0.25, so the language-centred comparison is limited to Russian and German, and the two nestings cannot be compared as complete codes of the five pairs at this threshold. Two effects make up the recall factor:

group, $\hat{D} \leq 0.25$	hierarchy, bits (k, β)	independent, bits	ratio
translation (ru, de)	1,029 (4, 3)	978	1.05
recall (ru, de, en)	24,972 (4, 1)	12,917	1.93
all five measurable cells	26,001	13,895	1.87
Russian (translation, recall)	3,363 (2, 3)	2,961	1.14
German (translation, recall)	3,682 (2, 3)	3,779	0.97
English (recall)	not reached	7,155	–

Table 3: Task-centred (top) and language-centred (bottom) hierarchies against the sum of the members' independent frontiers on the measurable cells, corrected protocol. The language groups were retrained on their two measurable members. The three-member language groups that include unmeasurable continuation cost 4,353 (ru) and 4,560 (de) bits as trained and, restricted post hoc to their measurable members, 1.45 and 1.19 times the independent programs; those values are indicative only, since their centres were trained with continuation in the group.

at matched (k, β) a hierarchy costs 0.94–1.5 times the flat runs with the same hyper-parameters, so the shared centre buys nothing, and the frontier comparison adds the cost of one β serving every member, since the only task-centred run in which all three languages reach 0.25 has a centre (14,600 bits) that alone costs more than the three independent programs together. Nor is either nesting cheaper than the other: at matched (k, β) the language-centred code costs 0.88–1.24 times the task-centred one over all eight cells (Table 6).

The per-member costs show no stable asymmetry. Given the translation centre (604 bits), a target language costs 200–225 bits, less than half of a standalone translation program; given a language centre retrained on the measurable members (770–975 bits), translation costs 137–139. Given the recall centre (14,600 bits), a language costs 3,200–3,950 bits, more than a standalone Russian or German recall program; given a language centre, recall costs 2,250–2,770, somewhat less than standalone, and for English no run reaches 0.25. Recall dominates every code that contains it, at 2,250–3,950 bits as a deviation from either kind of centre, and whichever group must carry it is the expensive one. The task given the language costs 0.6–0.7 times the language given the task with the retrained language groups and 0.9–1.3 times with the three-member ones; with one seed per hierarchy and up to 22% seed variation in the flat frontiers, neither direction is established, and in neither direction does a shared centre save bits.

Under the preregistered protocol the task-centred nesting had appeared 37% cheaper than the language-centred one and 11% cheaper than independent programs (Table 5). Both margins came from language-centred recall deviations of 5,300–26,200 bits, which the corrected protocol brings down to 3,200–3,400; they were selection artefacts of the kind described in Section 6, not properties of the model.

5.4 Composition

Six tied parts trained on every cell approach Design A at the same total slots on translation ($\hat{D}$ 0.017–0.027 against 0.005–0.007 at $2{+}2$) and cost about twice its distortion on recall (0.08–0.25 against 0.05–0.12). Assembled zero-shot for a held-out cell (Figure 4), the parts close 60–73% of the gap between no prefix and the floor on translation (three hold-outs over two cells, $\hat{D}$ 0.27–0.40) and 14–32% on recall (six hold-outs over three cells, $\hat{D}$ 0.68–0.86). Task and language parts compose partially for translation and barely for recall.

5.5 Distance between programs

Warm-starting from a prefix of the same task costs 0.85 of training from scratch; from the same language or an unrelated cell, 0.95 (Figure 5, left). The silhouette of the task partition on the symmetrised distance is 0.004 and that of the language partition -0.09: the direction is the hypothesised one, the structure is absent, and with seven pairs per group at one seed the difference is not significant. The zero-shot block (right) is the strong signal: a prefix applied to any other cell is roughly as bad as, or worse than, no prefix ($\hat{D}$ 0.7–1.4, median 0.97, up to 2.5), and a recall prefix applied to continuation

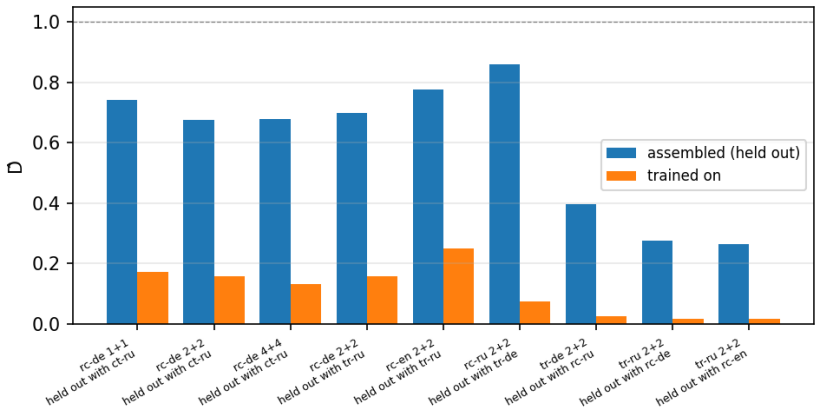


Figure 4: Assembled (held-out) against trained-on $\hat{D}$ of tied task and language parts, per hold-out.

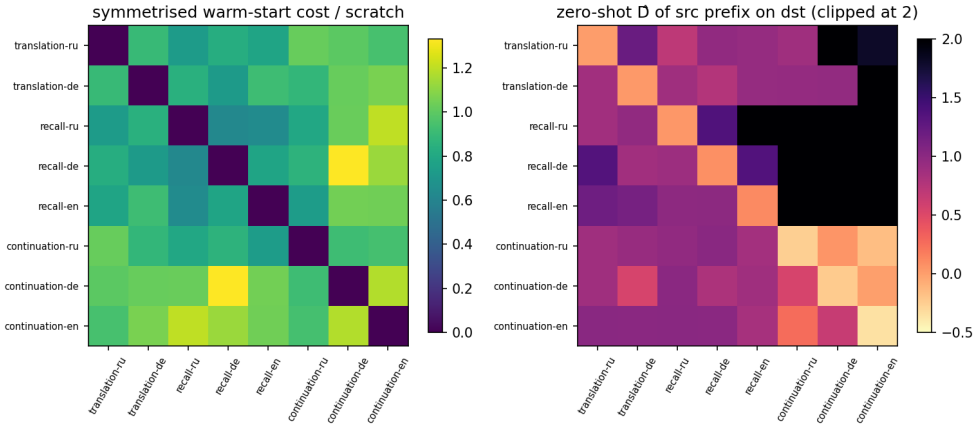


Figure 5: Left: symmetrised warm-start learning cost divided by the from-scratch cost. Right: $\hat{D}$ of a trained prefix applied unchanged to another cell, clipped at 2.

is catastrophic ($\hat{D}$ 5–25). The recall program overrides the model's default behaviour rather than adding to it.

5.6 Where the program is read

Patching block 0 alone recovers 70–87% of the effect for translation and 74–94% for continuation, but only 36–58% for recall, which has a secondary peak at block 17 (Figure 6). Single-block fractions summed over blocks come to 2.6 for translation and 3.0 for continuation, so their reading is redundant, but 1.3 for recall, which is close to additive. Supplying the block-0 attention input alone therefore reproduces most of a translation prefix's effect but only part of a recall prefix's, whose remaining effect is recovered at blocks 10–11 and 17. The probe measures where the effect can be recovered one block at a time; it does not show whether the prefix positions are attended later, nor what several blocks recover together.

Inside a hierarchical prefix the parts differ by depth profile rather than by peak, which is block 0 for every probe. The task centre is read early (on recall, its mass over blocks 0–3 is 1.0 and over blocks 14–27 it is 0.5); the language deviation on top of it is read early and again deep (1.6 and 1.9, with block 17 alone recovering 0.94 of its effect), and a task deviation under a language centre behaves like the task centre. This is a split by part in the opposite direction to the language-early, task-middle expectation drawn from Dumas et al. [3]. The recall centre without its English deviation is worse than a random prefix on English, so the recall centre is a non-English program. These probes were

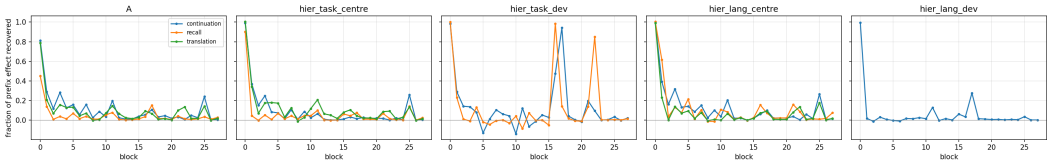


Figure 6: Fraction of a prefix's effect recovered by patching its positions into one block's attention input, mean over cells per task. Probes, labelled as in the figure: `A` is the Design A prefix at $k = 4$; `hier_task_centre` and `hier_task_dev` are the task centre alone and the language deviation on top of it; `hier_lang_centre` and `hier_lang_dev` are the language centre alone and the task deviation on top of it. Fractions are undefined where the donor is not at least 0.1 bits per token better than the recipient.

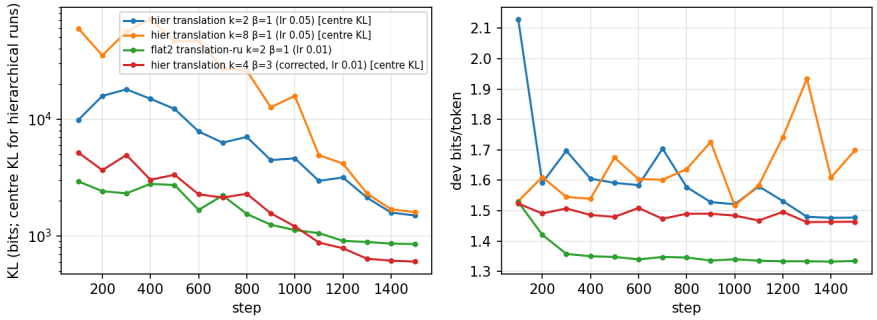


Figure 7: Centre KL (hierarchical runs) or total KL (flat run) and development loss against step for two preregistered-protocol hierarchical runs and two corrected-protocol runs (flat and hierarchical).

run on the preregistered hierarchies; on the corrected ones, probed at the posterior mean, the split replicates where the probe is defined. The probe was not run on posterior samples.

6 The objective is right, the search is hard

The saved histories of the 27 task-grouped hierarchical runs under the preregistered protocol show three failures (Figure 7). With learning rate 0.05, two prior standard deviations per Adam step, the centre KL overshoots early, peaking at a median of 26,000 bits at $k = 2$ and 144,000 at $k = 8$ before the cosine schedule lets the KL term pull the means back. The checkpoint rule, which selects on the distortion of the posterior mean, then picks a step where the rate is above its final value, by a median factor of 1.1–1.2 and up to 20. And the posterior mean is not the coded object: on translation at $\beta = 3$ the mean scores $\hat{D}$ 0.35–0.56 in three of four runs while samples score 0.06–0.17, because the model was trained on samples.

The corrected protocol removes all three. The centre KL of the 12 corrected hierarchical runs peaks 7–12 times lower, then falls monotonically, and the selected checkpoint is within 2% of the final rate. The measured frontier drops on every cell, by 22% (`translation-ru`, `recall-ru`), 58–63% (`translation-de`), 20% (`recall-de`) and 71% (`recall-en`), with four runs per cell instead of nine, and the rates become monotone in β. The protocol also decides a result: the preregistered 37% advantage of the task-centred nesting (Section 5.3) does not survive it. Shaw et al. [11] report that on a parity task the same objective reaches 870 bits from a hand-compiled initialisation but 2.8×10^6 bits from random initialisation, and leave the optimiser as the open problem. At this scale the gap closes mostly by selecting on the objective and stepping at the scale of the prior.

7 Limitations

Our description lengths depend on the frozen model, the datasets, the coding prior and the chosen distortion thresholds. They are not intrinsic Kolmogorov complexities of tasks or languages. In

particular, the KL-based cost depends on the diagonal Gaussian prior fitted to the model's token embeddings. Restricting the posterior to a mean-field form, with independent Gaussian coordinates, may also prevent shorter codes; the same restriction applies to the centre and deviation posteriors of the hierarchy. The reported bits are idealised expected bits-back coding costs, excluding coding overheads. Normalising distortion by the no-prefix loss is supported by the zero-rate baseline: the program $q = \pi_0$ performs only marginally better than no prefix (Appendix D). They also exclude the cost of choosing k and β: the frontier is the cheapest of four runs per cell whose held-out distortion is under the threshold, which is a mild selection on the test set with a small optimistic bias where a run sits close to the threshold. No preregistered point flips within that margin, but the exploratory 0.05 points are within one standard error of doing so.

Distortion is measured using teacher-forced negative log-likelihood. Since the model observes the correct preceding target tokens, it may infer the task or target language without relying entirely on the learned prefix. Thus, our measurements describe conditional prediction, not necessarily autonomous generation. Moreover, evaluation averages losses over posterior samples, with prefixes resampled across batches. This estimates expected distortion but does not guarantee that an individual sampled prefix will satisfy the threshold. We did not measure the distribution of full-test losses across individual prefix samples. The bootstrap intervals in Appendix E were computed using posterior-mean predictions, whereas the reported frontiers use posterior samples, so these intervals do not directly quantify uncertainty in the frontiers.

Two seeds were preregistered. The second seed of the corrected variational protocol reproduces the frontier within 4% on `translation-ru`, `recall-ru` and `recall-en`, and within 17–22% on `recall-de` and `translation-de`, in opposite directions, preserving the cell ordering (Table 8). The hierarchies and distance matrix use only one seed. The variational protocol was corrected after the initial experiments; both hierarchical groupings were rerun under the corrected protocol on the same grid ($k \in \{2, 4\}$, $\beta \in \{1, 3\}$, $\tau = 1$). The comparisons are therefore like for like, but rely on only four runs per group and one deviation-prior scale.

The training budget is limited to 1500 steps. Under the corrected protocol, all flat and hierarchical rates were still decreasing at the end of training, so the measured description lengths are upper bounds on the shortest achievable codes. The budget binds most for recall: a 3000-step run on `recall-ru` was still improving (Section 5.1). Checkpoint selection is also sensitive to development-loss fluctuations: one run (English centre, $k = 2$, $\beta = 1$) selected a transient loss reduction at step 500. Continuation has no meaningful measurable frontier because the frozen model already performs it nearly as well as the floor.

The mechanism experiments replace activations at the prefix positions in the self-attention input; intervening on the residual stream could produce different results. The corrected hierarchical mechanism probes were evaluated at posterior means rather than sampled prefixes, and some measurements were undefined. Finally, the exposure proxy needed to explain the language ordering was unavailable, so that comparison relies on only three languages.

8 Conclusion

Counting tokens misreads a prompt's complexity; counting bits under a bits-back code does not. At that currency we find translation programs of about 500 bits and Russian factual-recall programs of about 2,400 bits at the specified distortion thresholds, both upper bounds on the shortest achievable codes. Those bits are not organised as a cheap modifier on a shared core: neither nesting shows a consistent compression advantage across the tested configurations, and a centre plus deviations gives no consistent saving over independent programs. The asymmetry is mechanistic: recall programs are language-specific, do not compose, and are only partly recovered from the first block, whereas a translation program's effect is recovered almost entirely from the first block alone. The measurement is only as good as the optimiser, and the largest single effect in this study is the one obtained by selecting checkpoints on the objective rather than on distortion. Natural extensions are more languages stratified by resource level, larger models, a charge for the choice of k, the per-column mixture prior of Shaw et al. [11] as a second rate, a separate β for each member of a hierarchy, and more seeds.

Reproducibility The code, result tables, per-example losses, training histories and prefix checkpoints accompany this report; a single notebook regenerates every figure and table from the tables.

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A Preregistered comparisons

	hypothesis	outcome
H1	Within a task, the cost is smallest for English and grows as the language's share of the training data falls; within a language, it grows with task difficulty.	Task ordering supported (recall far above translation). Language ordering reversed on recall (English dearest), robust to the common items; no exposure measure was available.
H2	The language information in a prefix is small: giving the language saves few slots and bits, giving the task saves many.	Equal on the slot axis (1.8 slots each). In bits no stable asymmetry: task given language 0.9–1.3× language given task with three-member groups, 0.6–0.7× with retrained language groups, one seed each; the preregistered 1.3–7.8× asymmetry does not replicate.
H3	A task part and a language part compose at little cost for high-resource languages, with the cost growing as resource level falls; a shared centre saves bits.	Translation composes partially (60–73% of the gap zero-shot), recall barely (14–32%). No consistent saving from a shared centre: task-centred 1.87× independent programs on the measurable cells, retrained language groups 0.97–1.14×. The resource-level comparison was not run.
H4	The information-distance matrix over cells clusters by task, not by language.	Silhouette 0.004 (task) against -0.09 (language): the task direction, no structure, not significant.
H5	The language part of a prefix is read in early blocks and the task part in middle blocks [3].	Block 0 is the peak for both. The language deviation is read again at blocks 16–17 and the task centre is not: a split by part, in the opposite direction.
H6	Given the program, an input compresses to a similar number of vectors per sentence across languages and a different number per token.	One vector suffices with the task prefix in every cell but one. The per-token variant was not computed, and the vectors carry no rate term.

Table 4: Hypotheses fixed before the main runs, and the outcome of each comparison.

B Hierarchy: detailed comparisons

code for all eight cells, $\hat{D} \leq 0.25$	protocol	bits
centre per task + language deviations	prereg. selection	33,500
eight independent programs	prereg.	37,600
centre per language + task deviations	prereg. selection	53,500
centre per task + language deviations	corrected	26,700
centre per language + task deviations	corrected	not reached (ru, de: 8,900)
eight independent programs	corrected	17,600

Table 5: Three codes for the same eight cells under both protocols. The corrected independent row uses the corrected protocol on the five measurable cells and the preregistered runs on continuation; under the corrected protocol the language-centred hierarchy does not get `recall-en` below 0.25 within the grid, so only its Russian and German groups are summed.

(k, β)	task-centred, bits (worst $\hat{D}$)	language-centred, bits (worst $\hat{D}$)	ratio
(2, 1)	24,591 (0.28)	30,448 (0.59)	1.24
(2, 3)	12,693 (0.56)	11,182 (0.48)	0.88
(4, 1)	28,659 (0.23)	30,532 (0.28)	1.07
(4, 3)	11,180 (0.28)	13,396 (0.37)	1.20

Table 6: The two nestings at matched hyper-parameters under the corrected protocol: three centres plus eight deviations either way, summed over all eight cells, with the worst member distortion. At matched (k, β) a hierarchy costs 0.94–1.5 times the sum of the corrected flat runs with the same hyper-parameters; on recall the factor 1.93 of Table 3 decomposes as 1.12 (the matched excess at $(4, 1)$, 24,972 against 22,360) times 1.73 (one β for all members, 22,360 against 12,917). Under the preregistered rule a language given translation cost 485–662 bits and recall given a language 5,300–26,200.

C Input compressibility (exploratory)

With the $k = 4$ task prefix, one per-example vector replaces the input to within 10% of the real-input loss on every cell but `translation-ru`, which needs two; without the prefix, translation and continuation need two. There is no rate term on the vectors, so this is a slot count. On recall the bare model scores the target better with no input (3.1 bits per token) than with the real query (7.2), consistent with confident wrong answers once a subject is named.

D Zero-rate baselines

	draws from π_0		random token rows	
cell	$k = 2$	$k = 4$	$k = 2$	$k = 4$
translation-ru	0.96 (0.09)	0.91 (0.10)	0.97 (0.09)	0.96 (0.11)
translation-de	0.94 (0.04)	0.92 (0.04)	0.95 (0.06)	0.94 (0.09)
recall-ru	0.96 (0.04)	0.93 (0.06)	0.98 (0.04)	0.98 (0.08)
recall-de	0.98 (0.02)	0.97 (0.03)	0.98 (0.02)	1.00 (0.05)
recall-en	1.07 (0.05)	1.12 (0.06)	1.04 (0.04)	1.12 (0.12)

Table 7: $\hat{D}$ of untrained prefixes on the full test set, mean (sd) over eight draws. Draws from π_0 are the zero-rate program $q = \pi_0$; random rows of the token-embedding matrix are the recipient of the mechanism probes and are not rate 0. Neither improves on no prefix by more than a few percent of the gap, and on English recall both are worse than no prefix.

E Run inventory and replication

cell	seed 0, bits at 0.25 (k, β)	seed 1, bits at 0.25 (k, β)	ratio
`translation-ru`	542 (4, 3)	534 (4, 3)	0.99
`translation-de`	436 (2, 3)	533 (4, 3)	1.22
`recall-ru`	2,419 (2, 3)	2,486 (2, 3)	1.03
`recall-de`	3,343 (4, 3)	2,783 (2, 3)	0.83
`recall-en`	7,155 (4, 1)	6,850 (4, 1)	0.96

Table 8: Replication of the corrected-protocol frontier with a second seed; each seed's frontier is an independent replicate of the whole run grid. The binding level is 0.25 for `recall-de` and `recall-en` and 0.10 for the other cells, where the two levels coincide.

Design A, baselines and floors: 273 runs. Flat variational: 72 preregistered and 40 corrected runs (two seeds). Hierarchical: 54 preregistered and 24 corrected joint runs, plus 8 for the language groups retrained on their measurable members. Composition: 10 joint runs. Distance: 72 warm starts. Regime 2: 32 settings. Mechanism: 1,120 single-block patches, repeated on the corrected hierarchies at the posterior mean. One 3000-step recall run. Zero-rate baselines: 32 settings, eight draws each. Training: 1500 steps, batch 16, AdamW, learning rate 0.05 (0.01 under the corrected protocol), 50 warm-up steps, cosine decay, gradient clip 1.0, development evaluation every 100 steps; LoRA floors rank 16 on the attention projections, learning rate 2×10^{-4}, 1500 steps. Bootstrap intervals over test examples for the gain of the $\hat{D} \leq 0.10$ frontier runs over the no-prefix model, at the posterior mean: 0.52 ± 0.01 bits per token on `translation-ru`, 0.50 ± 0.03 on `translation-de` and 5.69 ± 0.07 on `recall-ru`.